

UNIVERSITÄT DES SAARLANDES

NNTI Winter Term Project 2025/2026
Language Identification from Audio

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1 Overview

In this project, you will build a spoken language identification system that classifies short audio segments into one of 22 Indian languages. In the current project setup, this is a challenging task due to limited training data and potential speaker bias within the dataset.

- **Task:** Audio Classification
- **Input:** Short speech segment from a single speaker
- **Output:** Language label (one of 22 languages)

An accurate and robust spoken language identification system is crucial for automated data collection pipelines and facilitate the development of multilingual applications such as voice assistants in call centers.

However, note that these constraints apply:

- Maximum model size is **600M parameters**
- **Multimodal LLMs** are not allowed
- No **external datasets** allowed

2 Background: Linguistic Diversity in India

India is one of the most linguistically diverse countries in the world, with thousands of languages and mother tongues spoken across the subcontinent. The Indian constitution recognizes 22 official languages, and many individuals within the society are multilingual, often speaking two or more languages in daily life. This diversity presents unique challenges and opportunities for speech technology; a robust language identification system must distinguish between closely related languages, handle code-switching, and account for regional variations. For more context, see [Languages of India](#) on Wikipedia.

3 Dataset

For this project, you will use a subset of a high-quality speech corpus covering 22 languages across the Indian subcontinent. You can download and explore the dataset on the Hugging Face hub [\[link\]](#).

Languages: Assamese, Bengali, Bodo, Dogri, Gujarati, Hindi, Kannada, Kashmiri, Konkani, Maithili, Malayalam, Manipuri, Marathi, Nepali, Odia, Punjabi, Sanskrit, Santali, Sindhi, Tamil, Telugu, and Urdu

3.1 Data constraints:

- 400 samples per language
- Only 5 speakers per language

4 Machine Learning Approach

To tackle this problem, you will adapt a pre-trained speech transformer for this classification task. The baseline uses supervised fine-tuning (SFT) of a multilingual variant of the wav2vec2 model, namely MMS-300m.

4.1 Baseline

The provided baseline fine-tunes [MMS-300m](#) on 22 languages.

- **Chance performance:** 4.54% top-1 accuracy
- **Baseline performance:** 31.96% top-1 accuracy
- **Training:** 3 epochs on L4 GPU (24 GB VRAM)

To help you begin the tasks, you are provided with a starter script `train_model.py` containing the baseline implementation. You may modify or extend this script as needed for your experiments.

5 Project Tasks

This project is divided into three tasks. In order to pass the project, you have to complete Task 1 and attempt task 2. For maximum point, you need to successfully complete all three tasks.

5.1 Task 1: Reproduce and Improve the Baseline

Your first task is to reproduce the baseline and improve its accuracy. In doing so, you will learn how to use the Hugging Face modules to process the dataset, train the model, and investigate the model performance. You can also try to improve the model performance as follows

- Experiment with different pre-trained models (e.g., w2v-BERT-2.0, etc.)
- Conduct better hyperparameter tuning (learning rate, batch size, warmup steps, etc.)
- Train longer or experiment with more careful training schedules

Take time to understand why this is not a simple classification problem like MNIST, since the acoustic similarities between related languages and the limited training data make this genuinely difficult.

Along with this document, you are provided with a starter script `train_model.py` to help you begin this task.

5.2 Task 2: Address Dataset Bias

The main challenge in the given dataset is that with only 5 speakers per language, the model may learn to identify speakers rather than languages. Your task is to prevent the model from learning the speaker identity as a shortcut to minimize the loss.

Possible approaches:

- **Data-centric:** Audio augmentation with pitch shifting, speed perturbation, or other spectral (frequency domain) or temporal (time-domain) transformations of the audio input [see [Audio augmentation in PyTorch](#)]. Moreover, neural approaches such as k -NN voice conversion might be effective [see [Baas et al., 2022](#)].

Model-based: Adversarial training to remove speaker information from learned representations [see, for example, Domain Adversarial Neural Networks (DANN) by [Ganin and Lempitsky, 2014](#)]. In the literature, adversarial neural networks have also been employed to remove protected attributes from text [[Elazar and Goldberg, 2018](#)].

- **Other:** Any other technique you can justify. This is open to your creativity.

5.3 Task 3: Analyze the Model

For this task, you will need to present an insightful analysis of the problem and your solution.

Possible analyses:

- Discuss the source of the bias in the data and explain why learning from biased data is difficult.
- Discuss your proposed technique and how it addresses the speaker bias problem. To what extent was the solution successful?
- Analyze the confusion patterns between languages. Which languages are most often confused with each other? why?
- Analyze the baseline and the improved model by visualization the representations of the last layer using the TSNE algorithm. Discuss whether or not the models encode the speaker identity.
- Any other analyses you find insightful for your model.

6 Instructions & Deliverables

6.1 Important Dates

- Code & Report Submission: **06.03.2026 (23:59)**
- Presentations: **11.03.2026 – 13.03.2026** and **18.03.2026 – 20.03.2026**

A sheet will be shared soon where you can select your preferred presentation slot.

6.2 Group Work & Submission Format

You are expected to work in a group of **2–3 students**.

Submit your project on **CMS** as a **single ZIP file**.

Naming convention: Name1_Name2_Name3.zip

The ZIP file must contain the following folders:

- **code/** — your implementation **requirements.txt**, and **README.md**)
- **report/** — your report sources and compiled PDF

Expected ZIP structure:

```
Name1_Name2_Name3.zip
  code/
    train_model.py
    requirements.txt
    README.md
    (any additional files/folders)
  report/
    main.tex (or your .tex entry file)
    main.pdf
    figures/
    (any additional assets)
```

6.3 Code Submission

Your code should be:

- Well-structured, readable, and thoroughly commented.
- Runnable on our systems without major modifications.
- Accompanied by a `requirements.txt` file listing all dependencies with versions.
- Documented in a `README.md` file that includes:
 - Python version used.
 - Clear instructions for running the code.
 - Any environment setup notes (e.g., Colab, local).

If you use code from public repositories, cite it clearly with URLs in comments.

6.4 Report Submission

Along with your code, submit a **4-page maximum report** (exclusive of references) in $\text{\LaTeX}$, using the [EACL 2023 Overleaf Template](#). Your report should:

- Cover all three project tasks, including methods, experiments, and analyses.
- Include citations for tools, libraries, and papers that informed your approach.

- Be included in the **report/** folder of the main submission ZIP and contain:
 - The **.tex** source file(s)
 - The compiled **.pdf**
 - All image/assets used

6.5 Presentation

Each team will present their work for **20 minutes** during the presentation dates (11–13 and 18–20 March 2026), followed by **5 minutes of Q&A**. All team members should contribute equally to both content and delivery. At least one member must be present in person.

6.6 Grading

Your final grade will be calculated as follows:

- **35%** – Completion of all required tasks (Tasks 1–3)
- **30%** – Depth of analysis and justification, supported by experiments and scholarly citations
- **20%** – Quality and clarity of presentation content
- **15%** – Delivery and engagement during the presentation

7 References

- [1] Wikipedia contributors. *Languages of India*. https://en.wikipedia.org/wiki/Languages_of_India. Accessed: *jaccess date*.
- [2] badrex. *nnti-dataset* (Hugging Face Datasets). <https://huggingface.co/datasets/badrex/nnti-dataset-full>. Accessed: *jaccess date*.
- [3] Meta AI (facebook). *facebook/mms-300m* (Hugging Face model card). <https://huggingface.co/facebook/mms-300m>. Accessed: *jaccess date*.
- [4] PyTorch / torchaudio. *Audio Data Augmentation Tutorial*. https://docs.pytorch.org/audio/stable/tutorials/audio_data_augmentation_tutorial.html. Accessed: *jaccess date*.
- [5] Matthew Baas, Benjamin van Niekirk, and Herman Kamper. *Voice Conversion With Just Nearest Neighbors*. In *Proc. Interspeech*, 2023. arXiv:2305.18975. <https://arxiv.org/abs/2305.18975>.
- [6] Yaroslav Ganin and Victor Lempitsky. *Unsupervised Domain Adaptation by Backpropagation*. arXiv:1409.7495, 2014. <https://arxiv.org/abs/1409.7495>.
- [7] Yanai Elazar and Yoav Goldberg. *Adversarial Removal of Demographic Attributes from Text Data*. arXiv:1808.06640, 2018. <https://arxiv.org/abs/1808.06640>.
- [8] Isabelle Augenstein and Andreas Vlachos. *EACL 2023 Proceedings Template* (Overleaf). <https://www.overleaf.com/latex/templates/eacl-2023-proceedings-template/fvjksksgsmnh>. Accessed: *jaccess date*.